

Polynomials

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Learning Outcomes

At the end of this lecture, you should be able to:

- ☐ Know what is a polynomial and identify the degree, coefficients, leading coefficient of a polynomial.
- ☐ Identify some special types of polynomials.
- ☐ Know what is meant by two polynomials are equal.
- ☐ Perform addition, subtraction, multiplication and substitution on polynomials.
- ☐ Know what is a root and factor of a polynomial.

Introduction

A **polynomial** is an expression of the form

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where n is a non-negative integer and $a_0, a_1, a_2, \dots, a_n$ are constants.

Let m be the largest integer such that $a_m \neq 0$, then

x : **indeterminate**

a_i : **coefficient** of x_i

m : **degree** of polynomial

a_m : **leading coefficient**

If all the coefficients are complex numbers, then the polynomial is called a **complex polynomial**.

If all the coefficients are real numbers, then the polynomial is called a **real polynomial**.

Polynomial of degree 0 are called **constant**.

Polynomial of degree 1 are called **linear**.

Polynomial of degree 2 are called **quadratic**.

Polynomial of degree 3 are called **cubic**.

Example

A polynomial p satisfies $p(x) = 0$ for all x if and only if all of its coefficients are zero. This unique polynomial is called the **zero polynomial**, and denoted by 0.

$$p(x) = 2x^2 + 3 \quad q(x) = ax^2 + bx + c \\ p(x) = q(x) \Leftrightarrow a=2, b=0, c=3$$

Equality: Two polynomials p and q satisfy $p(x) = q(x)$ for all x if and only if the corresponding coefficients of p and q are equal.

Example:

Polynomial	deg	a_m	Real/complex polynomial
$-3x^0$	0	-3	real (constant)
$8 + 6x$	1	6	real (linear)
$-9 + 2x + \sqrt{3}x^2$	2	$\sqrt{3}$	real (quadratic)
$1 + (-2)x + 0x^2 + 3x^3 + 4x^4$	4	4	real (quartic)
$4 + ix + 5x^2 + 7x^3$	3	7	complex (cubic)

leading coefficient
↓

Operations on Polynomials

Let $p(x) = a_0 + a_1x + \cdots + a_mx^m$ and $q(x) = b_0 + b_1x + \cdots + b_nx^n$ be two polynomials where $m \geq n$.

- **Addition:** $p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \cdots + (a_n + b_n)x^n + a_{n+1}x^{n+1} + \cdots + a_mx^m$.
- **Subtraction:** $p(x) - q(x) = (a_0 - b_0) + (a_1 - b_1)x + (a_2 - b_2)x^2 + \cdots + (a_n - b_n)x^n + a_{n+1}x^{n+1} + \cdots + a_mx^m$.
- **Multiplication:**
$$p(x)q(x) = (a_0 + a_1x + \cdots + a_mx^m)(b_0 + b_1x + \cdots + b_nx^n) = a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \cdots + (a_mb_n)x^{m+n}.$$

Remark: $\deg(p(x)q(x)) = \deg p(x) + \deg q(x)$
- **Substitution:** Let c be a fixed number, then
$$p(c) = a_0 + a_1c + \cdots + a_mc^m \text{ and } q(c) = b_0 + b_1c + \cdots + b_nc^n.$$

Operations on Polynomials

Example:

If $p(x) = 2x + x^2$ and $q(x) = 2 + 3x + x^3$, then

$$p(x) + q(x) = 2 + 5x + x^2 + x^3$$

$$p(x) - q(x) = -2 - x + x^2 - x^3$$

$$\begin{aligned} p(x)q(x) &= (2x + x^2)(2 + 3x + x^3) \\ &= 4x + 6x^2 + 2x^4 + 2x^2 + 3x^3 + x^5 \\ &= 4x + 8x^2 + 3x^3 + 2x^4 + x^5 \end{aligned}$$

$$p(2) = 2(2) + (2)^2 = 8$$

$$q(2) = 2 + 3(2) + (2)^3 = 16$$

Roots and Factors

A number a is a **root** (or **zero**) of a polynomial p if $p(a) = 0$.

The polynomials p_1 and p_2 are called the **factors** of a polynomial p if $p(x) = p_1(x)p_2(x)$ for all x .

Example:

Let $p(x) = x^3 + 1$.

$$\text{Since } p(-1) = (-1)^3 + 1 = 0$$

So $x = -1$ is a root of the polynomial p

$$p(x) = (x+1)(x^2 - x + 1) \text{ for all } x.$$

$x+1$ is a linear factor of p

and $x^2 - x + 1$ is a quadratic factor of p .